

IIM KOZHIKODE • MDP FOR IRMS OFFICERS • MAY 2026

SESSION 1

Predictive Safety & Risk Analytics

From Incidents to Prevention

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Head of AI Practice | Knorr-Bremse (2025-present)

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Lead Data Scientist | Mercedes-Benz (2008- 2021)

PhD in IEOR, IIT Bombay

A woman with long brown hair, wearing a yellow and white plaid shirt, is looking out of the open window of a train. The train is moving along tracks that lead into a vast, mountainous landscape. The mountains are covered in patches of snow and greenery, and the sky is overcast. The train's overhead power lines and tracks are visible in the foreground.

**"THE BEST WAY TO PREDICT THE FUTURE
IS TO CREATE IT."**

Driven to create the best solutions

Rail Vehicle Systems

- High-speed trains
- Regional & commuter trains
- Metros
- Light Rail Vehicles
- Monorails
- Locomotives
- Passenger cars
- Freight cars
- Off-train
- Special vehicles

Commercial Vehicle Systems

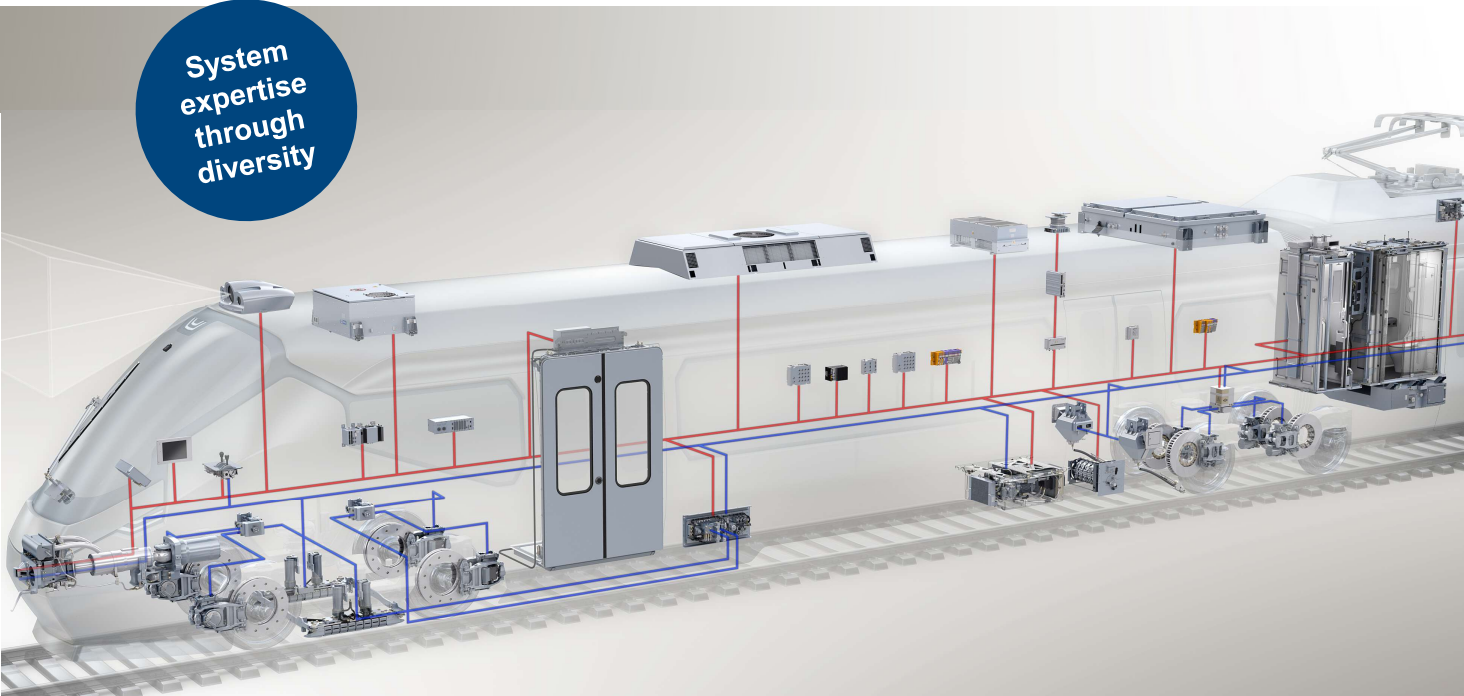
- Trucks
- Trailers
- Buses
- Engines
- Special vehicles



System expertise through diversity: braking systems, on-train & off-train products and service solutions over the entire vehicle life cycle



System
expertise
through
diversity



-  Braking Systems
-  Entrance Systems
-  Climate Control Systems / HVAC
-  Power Electronics
-  Rail Computing & Communication
-  Sanitary Systems
-  Wiper- / Wash Systems
-  Coupling Systems
-  Digital Solutions
-  Control, Command & Signaling

 KNORR-BREMSE  NEW YORK AIR BRAKE  IFE  MERAK  MICROELETTRICA  SELECTRON  EVAC  KB SIGNALING  ZELISKO  RAILSERVICES

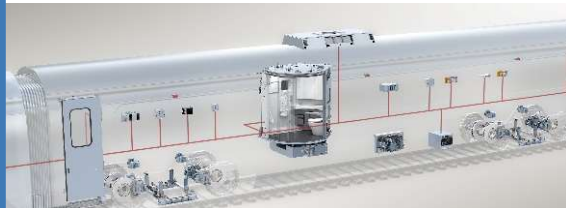
A variety of subsystems and additional services for all vehicle types



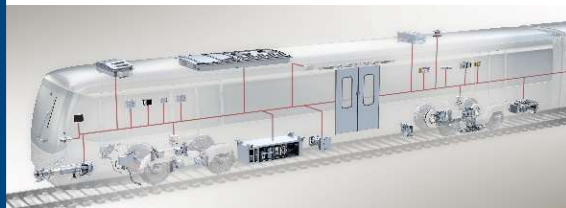
Freight cars



Passenger cars



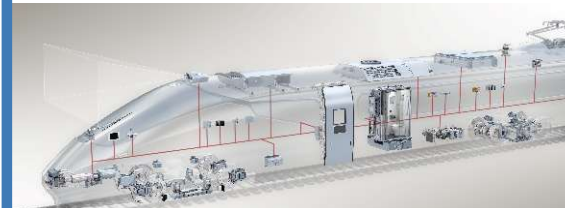
Metros



Locomotives



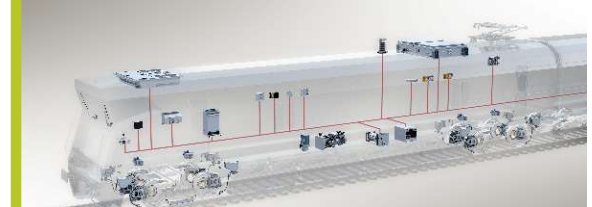
High-speed trains



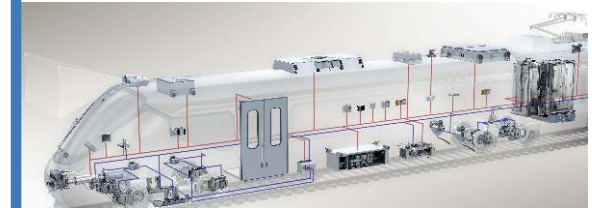
Light rail vehicles (LRV)



Special vehicles



Regional and commuter trains



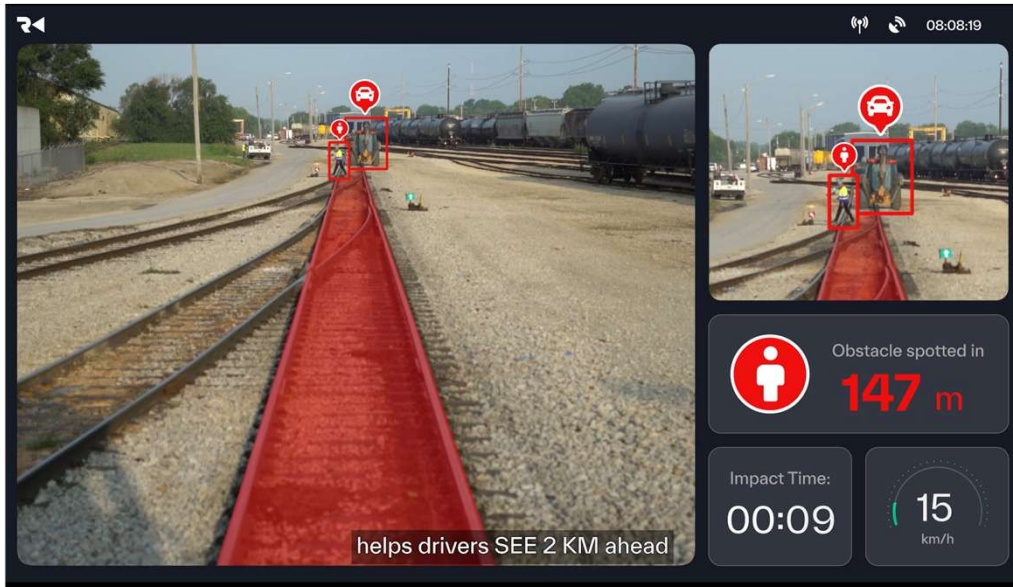
Monorails



— Freight transportation — Long-distance transportation — Urban transportation

Digital Services and group companies

Rail Vision



The most powerful hardware and software products for digitalisation in Rail



The Railster is Railnova's remote monitoring and edge device. It enables you to safely connect any type of rolling stock or component and stream data to your IT cloud in real time. The platform is easily updated to suit your needs at any time. The updatable software on the Railster allows you to perform edge computing and to compute your own rules, in order to receive custom fault codes and alerts.

[Discover >](#)



Railfleet is Railnova's digital ECM fleet and maintenance management platform. It enables you to view all your trains in one place and to communicate about fleet condition, availability and maintenance planning with

SESSION 1

Predictive Safety & Risk Analytics

From Incidents to Prevention

~90%

Drop in consequential
accidents: 135 (2014-15)
→ 14 (2025-26 to Feb)¹

75%

Of all consequential
accidents are derailments
(CAG 2022)²

₹1.3 L Cr

Proposed FY27 rail safety
outlay — largest ever³

¹ MoR via DD News (Mar 2026) ² CAG Report No. 22 of 2022 ³ Union Budget 2026-27

Session Roadmap — 3 Hours

AGENDA

SECTION 1

0–60 min

Safety Paradigm Shift & Rail Telematics

- › Reactive forensics vs. CBM
- › Diagnostic vs. predictive
- › Rolling stock & infrastructure telematics
- › Auto/CVS → heavy rail transfer

SECTION 2

60–135 min

Risk Modeling & Hands-On Exercise

- › Multivariate time-series data
- › Python notebook: anomaly detection
- › Predict signal degradation
- › Kaggle Bogie Vibration / CWRU

SECTION 3

135–180 min

Governance of Automated Safety

"Human-in-the-Loop" requirement

Decision framework for IRMS

Open discussion: high-prob events

Key takeaways & references

Key Takeaways from This Session

1

Understand the analytical shift from reactive forensics to proactive risk prevention using continuous telematics data.

2

Identify AI applications in safety-critical rail hardware and Condition-Based Monitoring (CBM).

3

Be able to process and interpret open-source sensor data to predict equipment degradation before a critical failure occurs.

SECTION 1

Moving to Condition-Based Monitoring • Rolling Stock & Infrastructure Telematics

CONSEQUENTIAL TRAIN ACCIDENTS

473

in 2000-01¹

→ 48

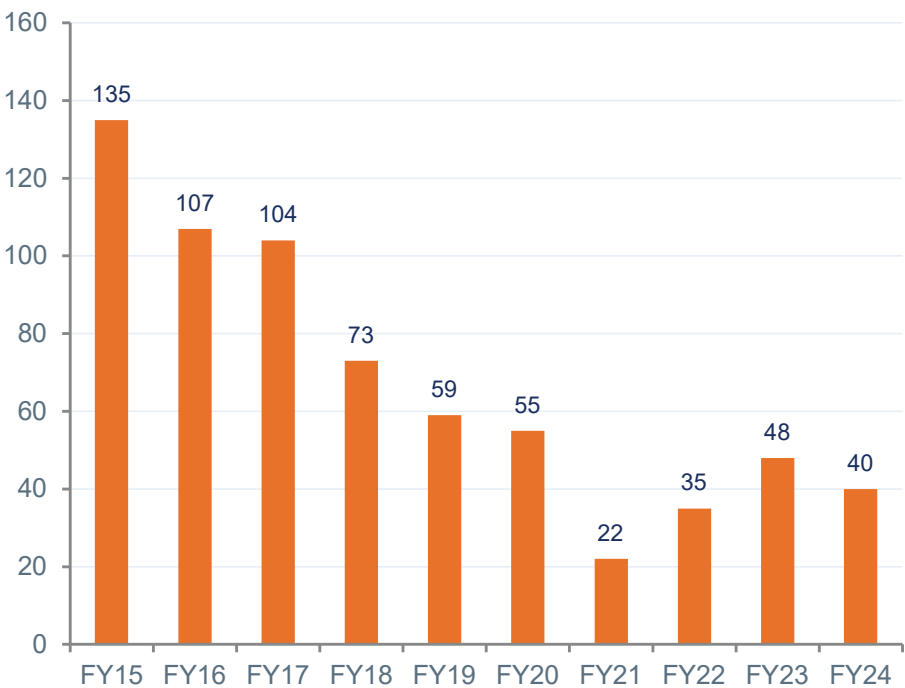
in 2022-23¹

→ 14

in 2025-26 (up to Feb 2026)²

~90% reduction in 10 years

Consequential Accidents — Annual Trend



Source: Open Government Data Platform India (Rajya Sabha Q.2195, 2024)⁴

[1] PIB Ministry of Railways, ID 1941440 (Jul 2023) [2] DD News citing MoR Lok Sabha reply (Mar 2026) [3] Metro Rail Today, FY24 safety data [4] data.gov.in / Rajya Sabha Q.2195 (2024)

The CAG Diagnosis — Why Derailments Persist

CAG 2022

75%

Of consequential accidents are derailments (2017-18 to 2020-21)¹

1,127

Total derailments studied; 289 (26%) linked to track renewals¹

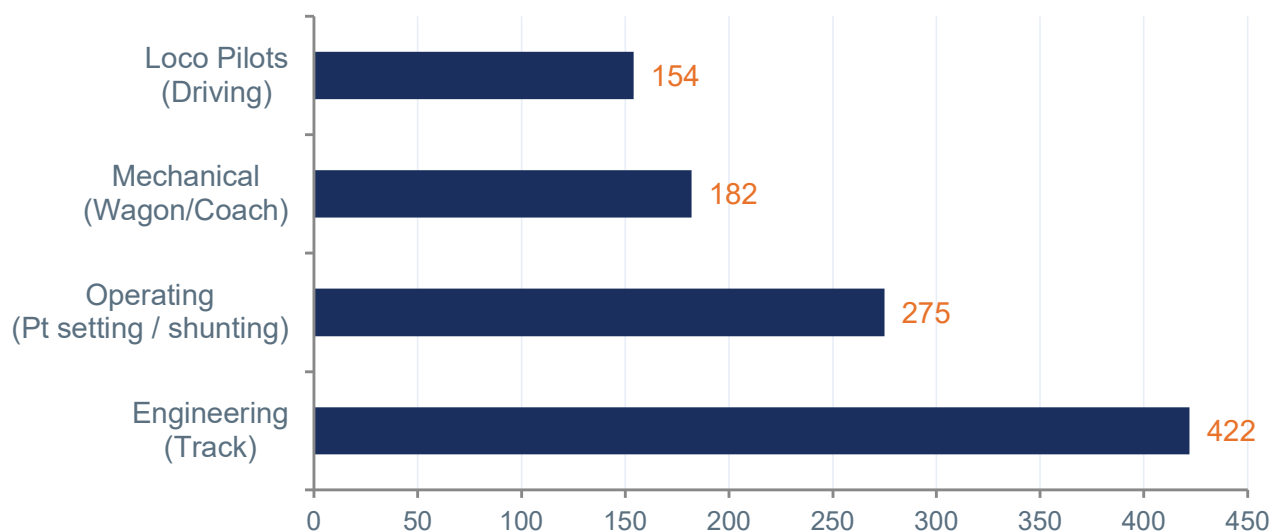
₹32.96 Cr

Damage to railway assets in sampled inquiries¹

30-100%

Shortfall in Track Recording Car inspections¹

Departmental Causes from 1,129 Inquiry Reports¹



KEY INSIGHT

Track + Mechanical = 53% of all derailments¹

These are exactly the failure modes that telematics + CBM can predict from continuous sensor data — vibration, temperature, geometry.

[1] CAG Report No. 22 of 2022 — Performance Audit on Derailment in Indian Railways (Union Government, Railways)

Case in Point — Balasore, 2 June 2023

CRS FINDINGS

BAHANAGA BAZAR

296

dead¹ • ~1,200 injured¹

128 km/h

Speed of Coromandel Express
entering the loop¹

*Avoidable with fused signalling +
interlocking telemetry analytics.*

CRS Probable Cause

Signalling-circuit alteration

ELB circuit of LC-79 (Balasore stn.) used to modify ELB circuit of LC-94 — practice deemed 'unsuitable' by CRS.

Aspect / Point mismatch

Home-signal aspect green for main line; crossover 17A/B set to the loop line.

No predictive fusion

Track-circuit, point-position and signal-aspect data were not analytically reconciled in real time.

[1] The Tribune (Jul 2023), citing Commissioner of Railway Safety findings [2] PMC12039713 — disaster-response analysis (peer-reviewed)

(a) Moving to CBM — Diagnostic vs. Predictive Analytics

Dimension	Reactive (Forensic / Post-Accident)	Predictive (CBM / Risk Analytics)
Trigger	Failure has occurred — CRS inquiry begins	Continuous sensor signal crosses risk threshold
Data Source	Incident reports, maintenance logs (offline)	Live telematics: GPS, IMU, acoustic, thermal sensors
Timeline	Post-event: hours to weeks for RCA	Real-time / near-real-time: seconds to minutes
Question	"What caused this failure?"	"What is the probability of failure in the next N hours?"
IR Example	Balasore CRS report (months)	WILD detecting wheel flat before next station
Output	Recommendations, FIR, asset replacement	Risk score → human-in-the-loop decision

Indian Institute of Technology Kanpur &
Research Designs & Standards Organisation (RDSO) Lucknow

WILD Wheel Impact Load Detection System



WILD - Online Wheel Impact and Flat Detection

A prototype automated system for On-Line estimation of Wheel Impact Loads and detection of Wheel Flats of running trains has been developed.

Identifying Precursor Events via Telematics

The shift: from post-accident analysis to detecting weak signals before they become incidents.

 GPS / GNSS	 Accelerometers (IMU)	 Acoustic / Vibration	 Thermal (IR)
PARAMETERS	PARAMETERS	PARAMETERS	PARAMETERS
Position, speed, heading Refresh: 1–10 Hz	3-axis g-force lateral, longitudinal, vertical	FFT spectrum, RMS amp. ISO 10816/20816 bands	Temperature (°C) vs. rolling baseline
DETECTS	DETECTS	DETECTS	DETECTS
Speed-section compliance, geofence breach, hunting at corners	Track geometry defects, derailment precursors, oscillation	Wheel flats, bearing defects, bogie instability	Hot axle bearings, brake pad overheating, electrical faults

[1] ISO 10816/20816 — Mechanical vibration evaluation [2] RDSO HAHW spec (rev. 2018) [3] LB Foster WILD-IV brochure (100 lb resolution)

(b) Rolling Stock Telematics — Real-time Hardware Health via ML

iCOM RAIL

Knorr-Bremse iCOM — What it monitors

Braking Systems

Brake cylinder pressure, pad wear,
anti-skid valve response, EP-CCB stack

CRITICAL

HVAC (Merak)

100,000+ units in service worldwide¹
Compressor health, refrigerant pressure

HIGH

Brake Control (Cube)

130,000+ CubeControl units in service¹
Valve actuation, pneumatic integrity

CRITICAL

Entrance Systems (IFE)

Door cycle counts, force profile,
obstacle-detect false-positive rate

MED

Vendor-projected outcomes (present as 'industry-claimed')

20%

Maintenance cost
reduction (iCOM)¹

2–4 yrs

ROI period for
iCOM deployment¹

120k+

New rail assets
connectable/year²

3,000+

Locos/MUs with
Railnova/Railster³

Figures are 'up to' vendor projections, not audited operator KPIs.

[1] Knorr-Bremse / OpenText case (Seckler, 2017) [2] Knorr-Bremse + Nexxiot newsroom (Dec 2023) [3] Knorr-Bremse Railnova story

How OEMs Apply ML to Safety-Critical Rail Hardware

Knorr-Bremse iCOM Rail

20% maintenance
cost reduction¹

2–4 yr ROI¹

200k+ connected
freight assets via Nexxiot²

¹ OpenText case 2017 ² KB newsroom

Alstom HealthHub

Up to 20%
material savings³

Up to 30% reduction
in train downtime³

Up to 50% reduction
in recurring faults³

³ Alstom HealthHub product pages

Siemens Railigent X

Up to 40% reduction
in delay costs⁴

Up to 30% fewer
unscheduled depot stops⁴

Up to 15% lower
maintenance costs⁴

⁴ Siemens Mobility product pages

[1] All figures are 'up to' vendor projections, not audited operator KPIs — present accordingly. Sources: OpenText 2017; Alstom HealthHub & FlexCare pages; Siemens Mobility Railigent X.

Translating CVS / Automotive ML Expertise → Heavy Rail

Automotive / CVS Origin	→	Heavy Rail Application	Adaptation Needed
ABS wheel-speed sensor	→	Wheel Impact Load Detector (WILD) — 100 lb resolution ²	Higher mass, kN-scale impact loads; ruggedised enclosure
EV battery thermal camera	→	Hot Axle Box Detector (HABD) — RDSO-SPN-RE-HAHW 2018 ¹	Wayside deployment; speed-compensated thermal calibration
EV motor vibration monitoring	→	Bogie IMU array (CWRU-style bearing diagnostics) ³	Multi-axis fusion; ISO 10816 / 20816 evaluation bands
OBD-II engine fault codes	→	Locomotive TCMS / iCOM event logging	Multi-vendor TCMS integration; IR-domain protocols
ADAS camera fusion	→	Forward-vision OHE / track inspection	25 kV EMI shielding; real-time LiDAR fusion at speed

[1] RDSO Hot Axlebox/Hot-Cold Wheel Detector Spec (RDSO-SPN-RE-HAHW-2012, rev. 2018) [2] LB Foster WILD-IV brochure (100 lb resolution) [3] CWRU Bearing Dataset (Case Western Reserve University) — de facto PdM benchmark

Global Benchmarks — Where Are the Targets?

EU-27 (2023)

1,565

*significant railway
accidents¹*

841 fatalities; 650 serious injuries.
Total accident cost ≈ €3.5 bn/year¹

UK (2024-25)

207

*mainline train
accidents²*

14 non-workforce fatalities;
29 high-potential incidents²

–40%

*Derailment rate
since 2005³*

2024 = 2nd-lowest year on record;
train-accident rate –33% since 2005³

India's challenge: ~7,300 stations, ~68,000 RKM, ~13,000 trains/day — at scale that exceeds EU + UK + US combined.

[1] ERA Safety Overview 2025; Eurostat [2] UK ORR Data Portal — Rail Safety 2024-25 [3] Association of American Railroads, citing FRA Form 6180.54 data (2025)

SECTION 1 — CHECKPOINT

Q1

Which of the four sensor types (GPS, IMU, acoustic, thermal) would you deploy first in your zone — and why? Justify in terms of dominant failure modes from the CAG 2022 break-up.

Q2

If 53% of derailments are Track + Mechanical (CAG 2022), what continuous data streams already exist on Indian Railways that we are NOT yet analytically fusing?

Q3

The Balasore CRS report cited a signalling-circuit alteration. What would a predictive risk model need as input to flag this BEFORE the train enters the loop?

Think-Pair-Share (3 min) → Class debrief (2 min)

SECTION 2

(a) Exploring Railway Telematics Datasets • (b) Predictive Maintenance with Sensor Data

75 MINUTES

(a) Multivariate Time-Series Data — What It Looks Like

Properties of Rail Sensor Streams

High frequency

Vibration 1–20 kHz; GPS 1–10 Hz; thermal 0.1–10 Hz

Multivariate & correlated

Many sensors per asset; cross-channel dependencies

Non-stationary

Statistics drift with speed, load, track profile, weather

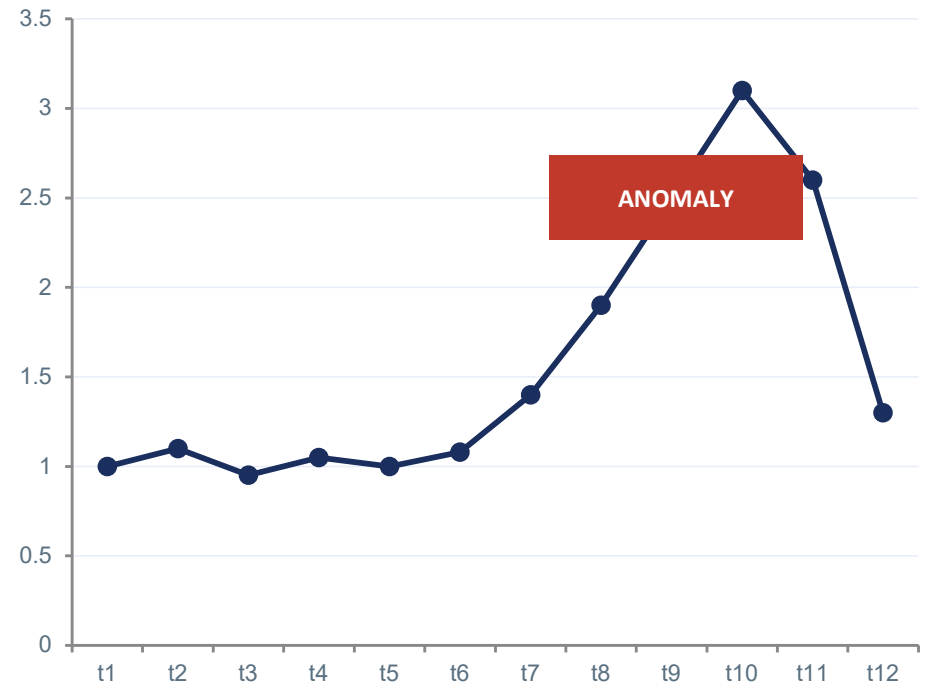
Noisy

EMI, mechanical coupling, sensor drift, calibration loss

Sparsely labelled

Faults are rare → unsupervised methods dominate

Conceptual: Vibration Amplitude over Time



Goal: detect the rising trend at t6–t7 — BEFORE peak failure at t10.

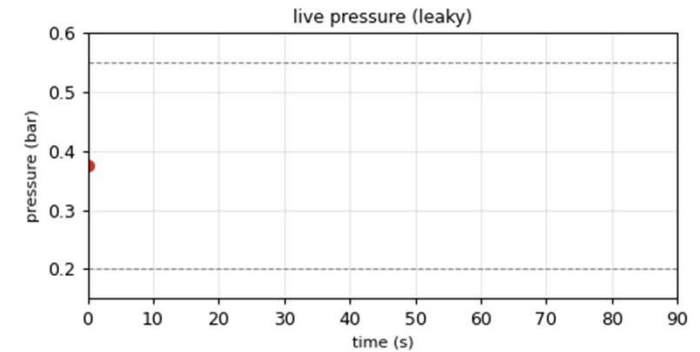
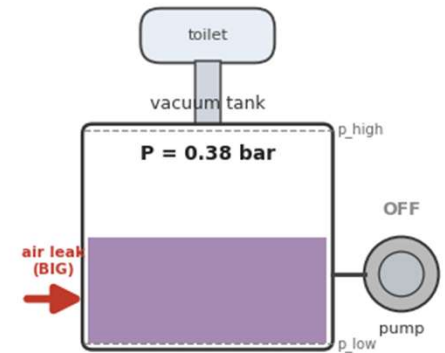
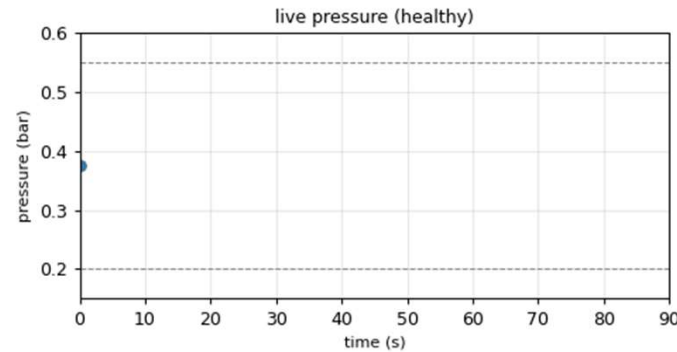
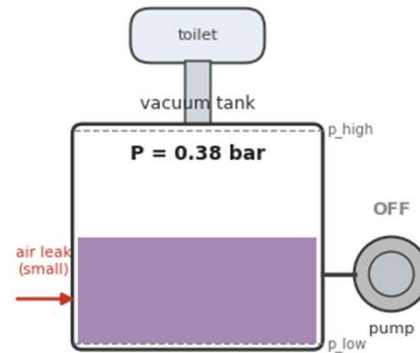
Industry-Relevant Tutorial (Anomaly Detection Basics)

Leak detection



https://github.com/meettalak/IIMK-Tutorials/blob/main/railway_toilet_leak_detection.ipynb

Railway toilet vacuum system — healthy vs leaky
Healthy toilet — small leak **Leaky toilet — big leak**

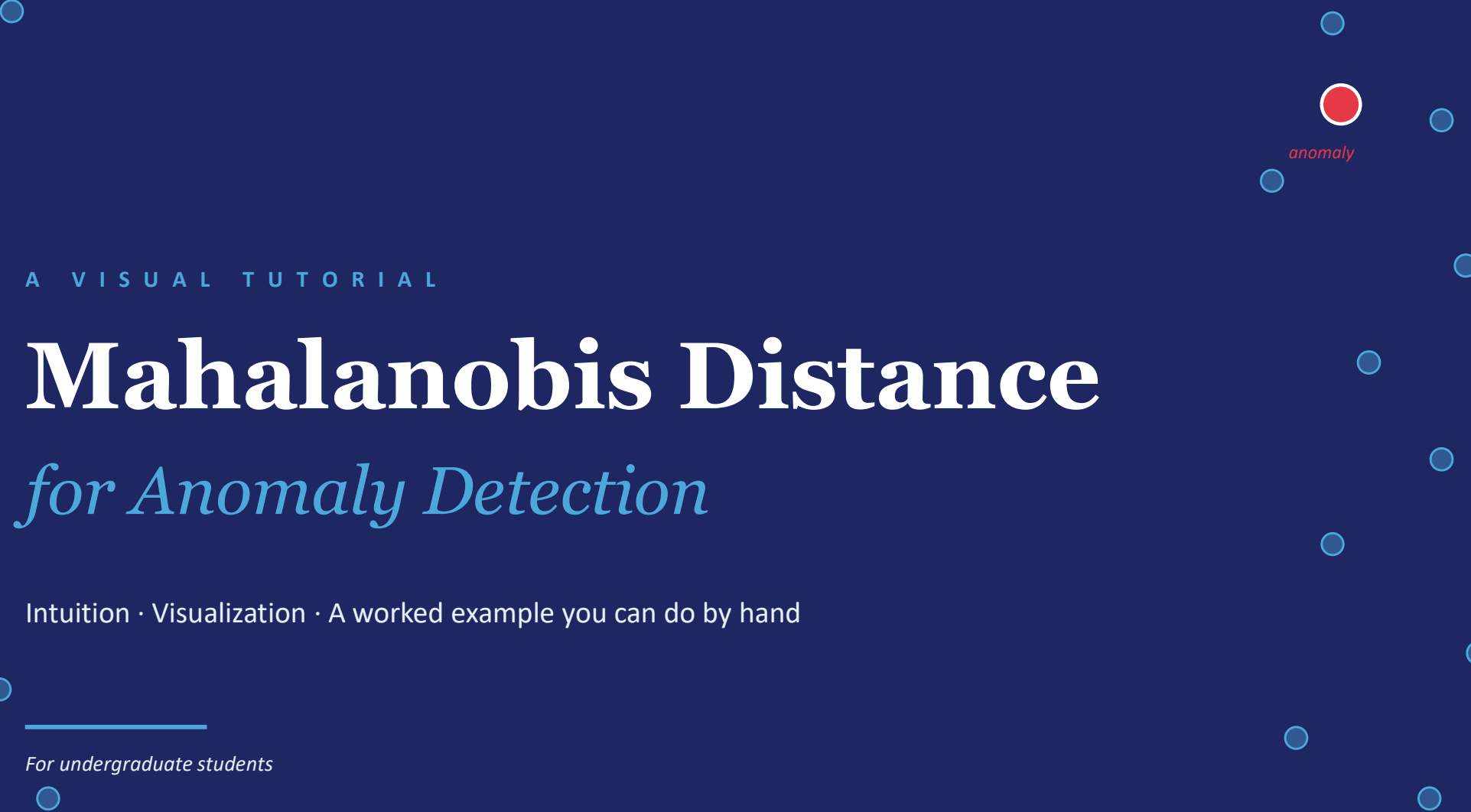


Quiz?



Who is he?





A VISUAL TUTORIAL

Mahalanobis Distance

for Anomaly Detection

Intuition · Visualization · A worked example you can do by hand

For undergraduate students

THE PROBLEM

What is Anomaly Detection?

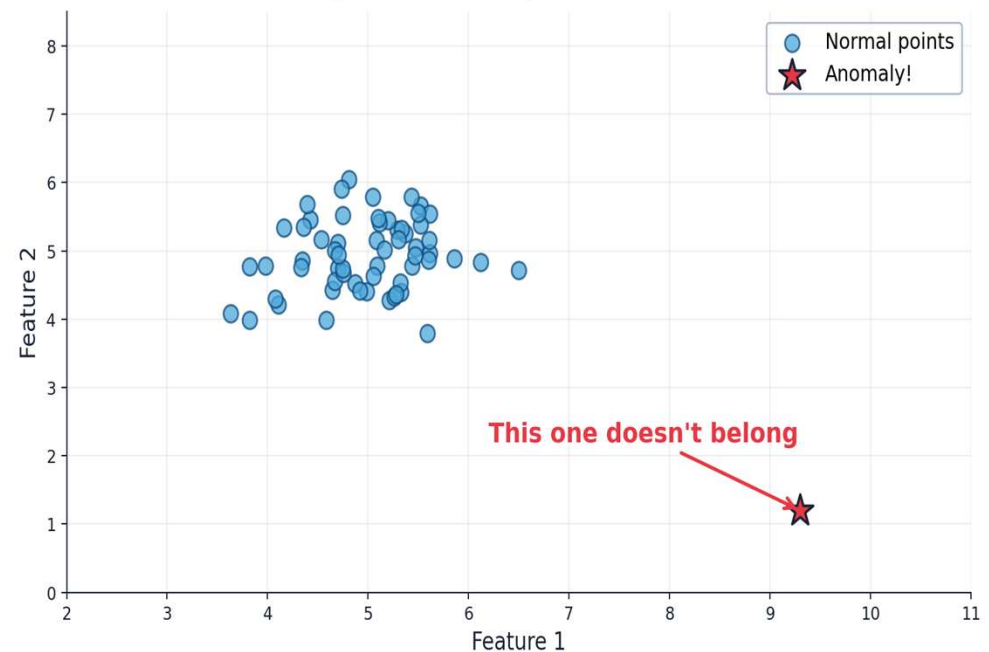
Find the data points that look different from everyone else.

We learn what "normal" looks like from a sample of data, then flag anything that doesn't fit that pattern.

Where it shows up

- Fraud detection — a transaction that's nothing like your usual spending
- Manufacturing — a sensor reading that signals a defective part
- Medicine — a lab result that looks unusual versus healthy patients
- Cybersecurity — network traffic that breaks from normal patterns

Anomaly Detection: Spot the Odd One Out



60 "normal" customers and one suspicious one. The eye spots it instantly — but how do we teach a computer to?

First Instinct: Just Measure Distance

"Far from the average = weird."

This is the Euclidean distance — straight-line distance from the center (mean) of the data.

Euclidean formula (2D)

$$D = \sqrt{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2}$$

x = the point we're testing

μ = the mean (center) of the data

D = how many units away

The rule

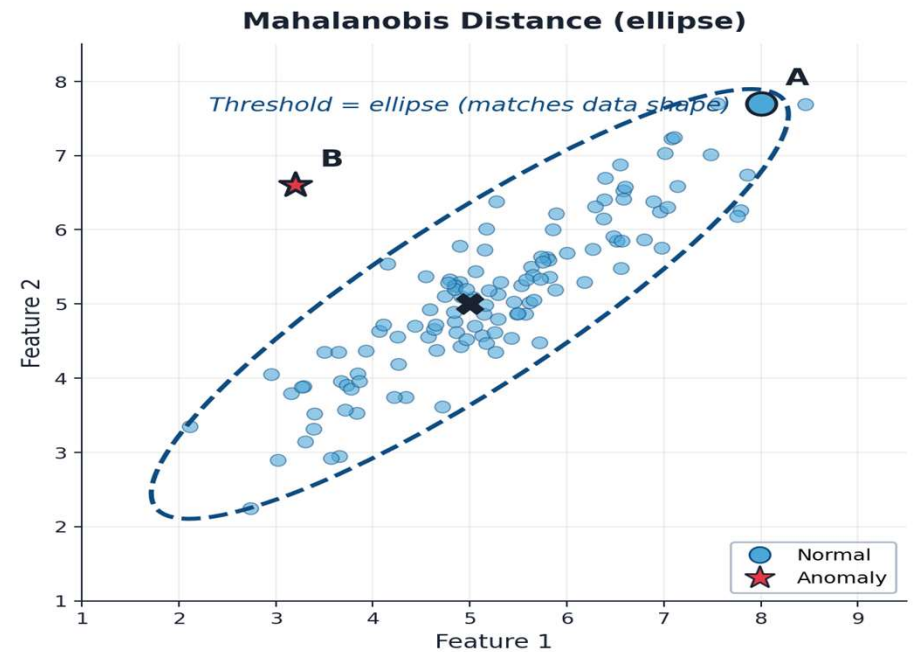
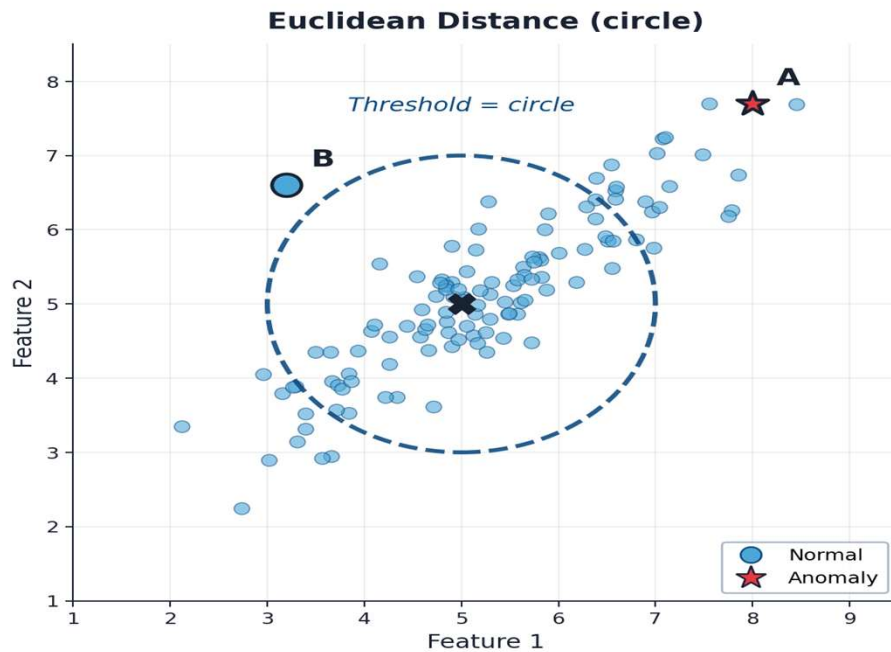
If D is bigger than some threshold T , call the point an anomaly.

*Visually: this draws a circle around the mean. Inside the circle = normal.
Outside = suspicious.*

⚠ Simple — but it can be very wrong. See next slide.

But Distance Alone Isn't Enough

Same data, two distance rules — different verdicts



Left (Euclidean): Point A is flagged because it's far. Point B is accepted because it's close. **Both verdicts are wrong.**

Right (Mahalanobis): A fits the data's tilted shape → normal. B breaks the pattern → **correctly flagged.**

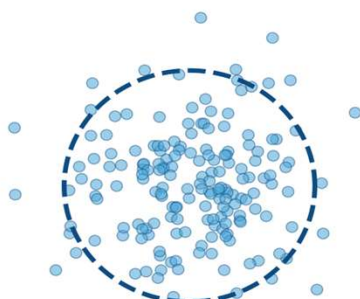
THE KEY INSIGHT

Data Has a Shape — and Direction Matters

Real datasets are rarely perfect circles. Features have different spreads and often correlate with each other.

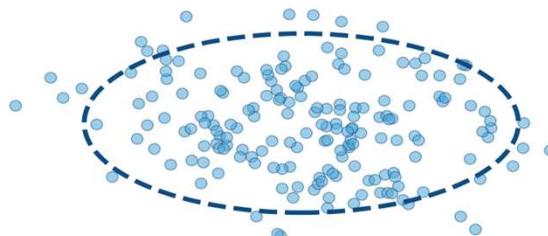
The covariance matrix encodes the data's shape

Circular cloud



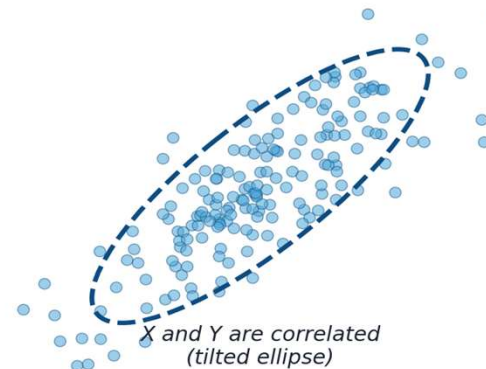
*X and Y vary equally,
no correlation*

Stretched cloud



*X varies more than Y,
no correlation*

Tilted cloud



*X and Y are correlated
(tilted ellipse)*

Big idea: the **covariance matrix** Σ captures this shape — and Mahalanobis distance uses it.

Enter Mahalanobis Distance

"How many standard deviations away is this point — along the data's own axes?"

In plain English: stretch and rotate space until the data cloud becomes a perfect circle. Then measure distance.



Accounts for spread

A feature that varies a lot needs more deviation before raising alarm than a tight one does.



Accounts for correlation

If two features move together, a point that breaks the pattern is more suspicious than one along it.



Scale-free

Units don't matter — the same answer whether you measure in cm or meters, dollars or rupees.

Next: the formula — and how it does this with simple matrix algebra.

DON'T PANIC

The Formula, Decoded

$$D(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \cdot \mathbf{S}^{-1} \cdot (\mathbf{x} - \boldsymbol{\mu})}$$

$$\mathbf{x} - \boldsymbol{\mu}$$

How far from the center?

Take the point minus the mean. This is a vector — the offset from the data's center.

$$\mathbf{S}^{-1}$$

The shape correction

The inverse covariance matrix. It stretches and rotates so the data cloud becomes a unit circle.

$$(\dots)^T \cdot \mathbf{S}^{-1} \cdot (\dots)$$

Scaled, rotated distance²

Multiply it all together. Take the square root. You get distance in "standard-deviation units".

Let's Compute It by Hand

1 Set up the data and compute the mean

Our 5 training points

Point	x_1	x_2
P_1	7	6
P_2	3	4
P_3	6	6
P_4	4	4
P_5	5	5
Sum	25	25

Step 1a — compute the mean μ

$$\mu_1 = (7 + 3 + 6 + 4 + 5) / 5 = 25 / 5 = 5$$

$$\mu_2 = (6 + 4 + 6 + 4 + 5) / 5 = 25 / 5 = 5$$

Mean vector $\mu = (5, 5)$

Step 1b — compute deviations $(x - \mu)$ for each point

$$\begin{aligned} P_1: (7-5, 6-5) &= (2, 1) & \bullet & P_2: (3-5, 4-5) = (-2, -1) & \bullet & P_3: (6-5, 6-5) = (1, 1) \\ P_4: (4-5, 4-5) &= (-1, -1) & \bullet & P_5: (5-5, 5-5) = (0, 0) \end{aligned}$$

Build the Covariance Matrix S

2 Measure spread and correlation between features

Using sample formulas (divide by $n - 1 = 4$):

$$\text{Var}(X_1) = \Sigma(x_1 - \mu_1)^2 / 4 = (4 + 4 + 1 + 1 + 0) / 4 = \mathbf{10 / 4 = 2.5}$$

$$\text{Var}(X_2) = \Sigma(x_2 - \mu_2)^2 / 4 = (1 + 1 + 1 + 1 + 0) / 4 = \mathbf{4 / 4 = 1.0}$$

$$\text{Cov}(X_1, X_2) = \Sigma(x_1 - \mu_1)(x_2 - \mu_2) / 4 = (2 + 2 + 1 + 1 + 0) / 4 = \mathbf{6 / 4 = 1.5}$$

Assemble the covariance matrix S:

$$S = \begin{bmatrix} 2.5 & 1.5 \\ 1.5 & 1.0 \end{bmatrix} \begin{matrix} \leftarrow \text{Var}(X_1), \text{Cov} \\ \leftarrow \text{Cov}, \text{Var}(X_2) \end{matrix}$$

Symmetric matrix: $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$

Invert the Covariance Matrix

3 Compute S^{-1} — the "shape correction"

Recipe for a 2×2 inverse

If $S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, then:

$$S^{-1} = (1 / \det) \times \begin{bmatrix} c & -b \\ -b & a \end{bmatrix}$$

where $\det = a \cdot c - b^2$

For our S ($a=2.5$, $b=1.5$, $c=1$):

$$\det = (2.5)(1) - (1.5)(1.5) = 2.5 - 2.25 = \mathbf{0.25}$$

$$1 / \det = 1 / 0.25 = 4$$

Apply the recipe

$$S^{-1} = 4 \times \begin{bmatrix} 1.0 & -1.5 \\ -1.5 & 2.5 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} 4 & -6 \\ -6 & 10 \end{bmatrix}$$

Clean integer values — we're ready!

Quick check: $S \cdot S^{-1}$ should give the identity matrix. Multiplying $\begin{bmatrix} 2.5 & 1.5 \\ 1.5 & 1 \end{bmatrix} \times \begin{bmatrix} 4 & -6 \\ -6 & 10 \end{bmatrix}$ gives $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. ✓

Plug In a Test Point and Get the Distance

4 Test point $x = (3, 8) \rightarrow$ compute $D(x)$

The computation, line by line

① Deviation from the mean:

$$x - \mu = (3 - 5, 8 - 5) = (-2, 3)$$

② Multiply $S^{-1} \times (x - \mu)$:

$$\begin{bmatrix} 4 & -6 \\ -6 & 10 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} (4)(-2) + (-6)(3) \\ (-6)(-2) + (10)(3) \end{bmatrix} = \begin{bmatrix} -26 \\ 42 \end{bmatrix}$$

③ Dot with $(x - \mu)$:

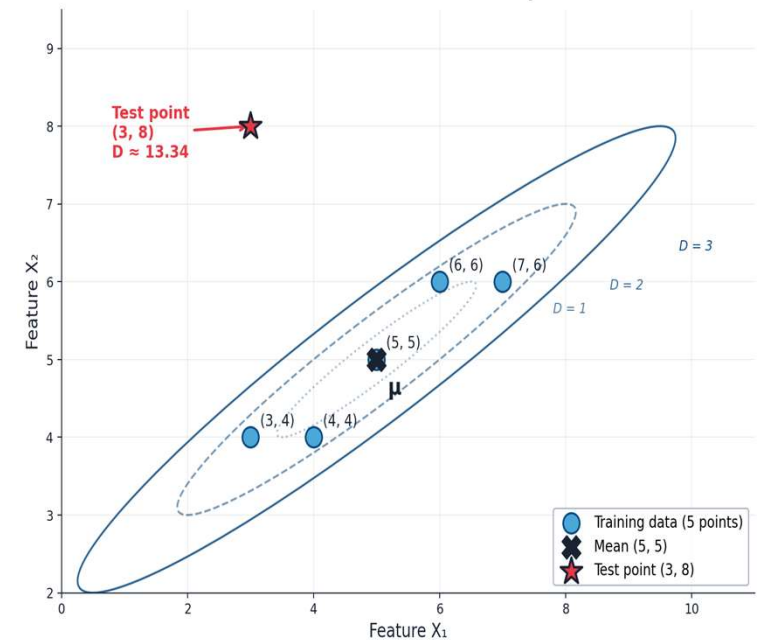
$$(-2, 3) \cdot (-26, 42) = (-2)(-26) + (3)(42) = 52 + 126 = 178$$

④ Take the square root:

$$D = \sqrt{178} \approx \mathbf{13.34}$$

This is the Mahalanobis distance of the test point from the training data.

Our Hand-Calculation Example



PUTTING IT ALL TOGETHER

From Distance to Anomaly Verdict

Pick a threshold T . Flag the point if $D > T$.

A common rule of thumb for 2-feature data: threshold $\approx \sqrt{5.99} \approx 2.45$ (95% confidence under a Gaussian assumption).

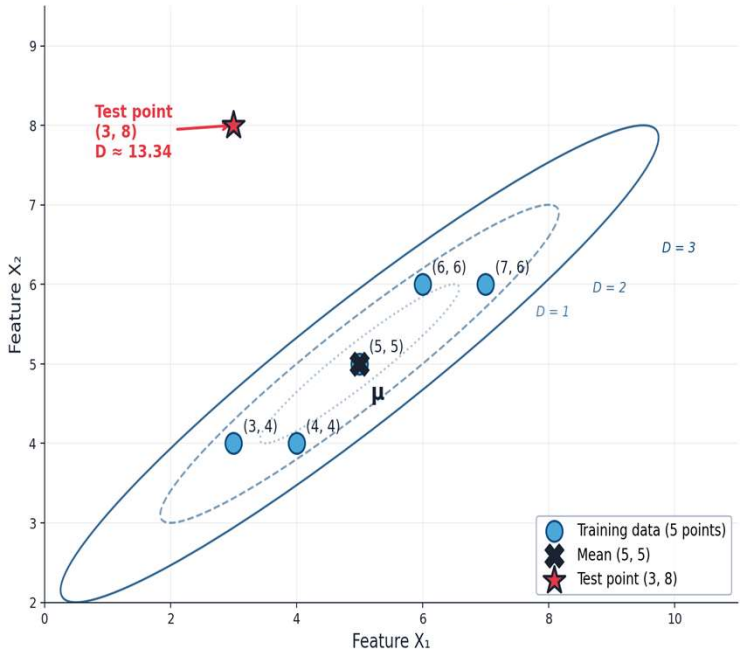
How our two test points compare

Point	Euclidean D	Mahalanobis D	Verdict
(7, 6)	≈ 2.24	≈ 1.41	✓ Normal
(3, 8)	≈ 3.61	≈ 13.34	✗ Anomaly!

Why the gap is so big

Euclidean only sees (3, 8) as 1.6× farther than (7, 6). Mahalanobis sees it as ≈9.4× farther — because (3, 8) breaks the positive correlation in the data, while (7, 6) follows it.

Our Hand-Calculation Example



Solid line: $D = 3$ ($\approx 99\%$ confidence). Test point sits far outside.

KEY TAKEAWAYS

What you learned today

1

Euclidean = circle, Mahalanobis = ellipse

Euclidean draws an identical circle around the mean. Mahalanobis adapts its boundary to the data's spread and tilt.

2

The covariance matrix tells the shape

Diagonal entries = how much each feature varies. Off-diagonal = how features move together. S^{-1} undoes that shape.

3

Distance is now in standard-deviation units

A Mahalanobis distance of 3 means "3 standard deviations along the data's natural axes." Larger D = stranger point.

4

Used everywhere — fraud, defects, outliers

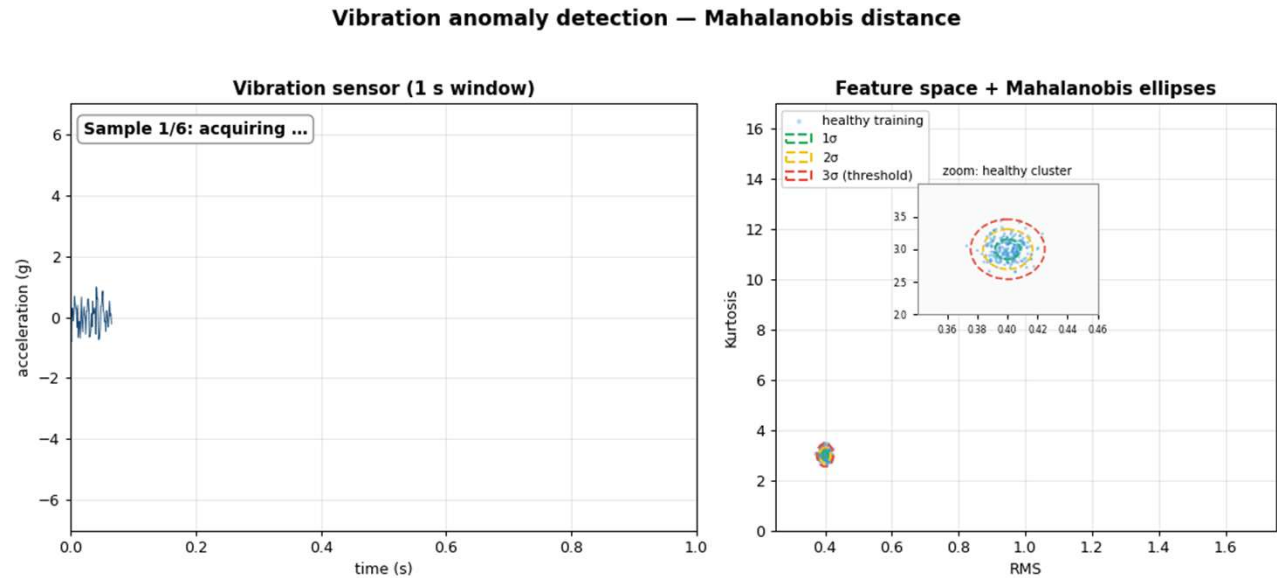
Foundation of multivariate outlier detection, Hotelling's T^2 , linear discriminant analysis, and many Gaussian-based methods.

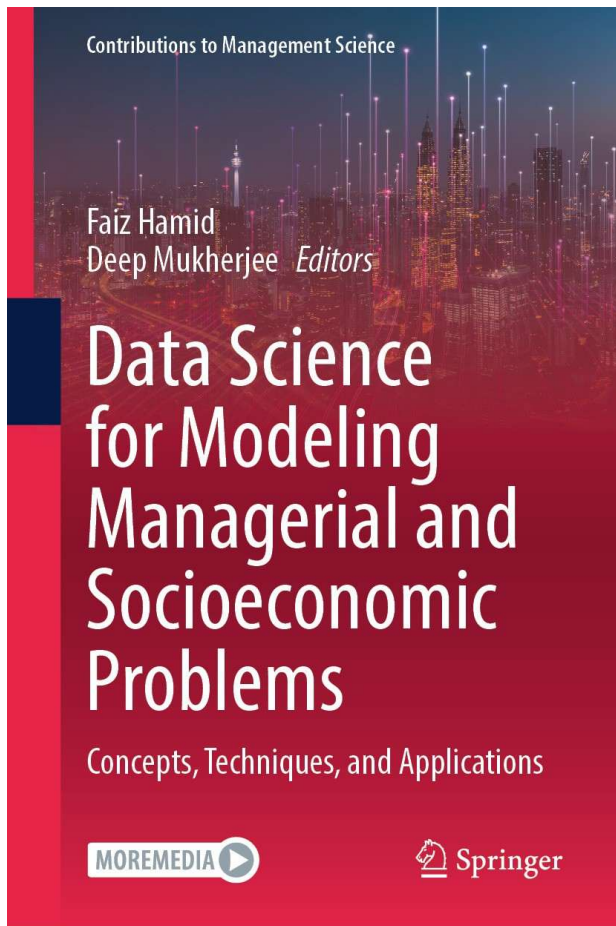
PUTTING IT ALL TOGETHER

Example 2: Vibration signal analysis



https://github.com/meetilak/IIMK-Tutorials/blob/main/Mahalanobis_Vibration_Anomaly_Detection.ipynb





Condition Monitoring for Shinkansen Bogies Based on Vibration Analysis *

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Thakkar, K., Hemachandra, N., Singh, T.R. (2026). **Anomaly Detection Methods: Application to Automated Vehicle Health Monitoring**. In: Hamid, F., Mukherjee, D. (eds) *Data Science for Modeling Managerial and Socioeconomic Problems*. Contributions to Management Science. Springer, Singapore.
https://doi.org/10.1007/978-981-97-9060-9_14

CWRU Bearing Dataset — Case Western Reserve University¹

EDM-seeded faults at 0.007"—0.040" on inner race, ball, outer race • 12 kHz & 48 kHz sampling • Motor loads 0–3 HP (1797–1720 rpm) •
The de-facto Predictive-Maintenance benchmark used in 1,000+ peer-reviewed papers

Kaggle Bogie Vibration

Simulated multivariate IMU time-series
across normal and bogie-fault states
(air-spring, wheelset, yaw, etc.).

[kaggle.com](#) (search 'bogie vibration')

Real-Time Railway Signaling Fault

Kaggle dataset of signaling-fault events
— useful for classification of signal-
circuit anomalies.

[kaggle.com](#) (signaling fault)

NASA Prognostics C-MAPSS

Turbofan & bearing prognostics
datasets — common transfer baseline
for rail PdM models.

[nasa.gov/intelligent-systems-division](#)

[1] CWRU Bearing Data Center, Case School of Engineering. <https://engineering.case.edu/bearingdatacenter> — IEEE bearing-fault review: 10.1109/ACCESS.2020.2988293

(b) Hands-On: Predictive Maintenance with Sensor Data — Setup

STEP 1

```
# — Cell 1: Setup —————
import pandas as pd, numpy as np, matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import IsolationForest
from scipy.io import loadmat
from scipy.stats import kurtosis, skew

# — Cell 2: Load CWRU bearing dataset —————
# Source: https://engineering.case.edu/bearingdatacenter
data = loadmat('48k_Drive_End_OR007@6_0_130.mat')
signal_normal = data['X097_DE_time'].ravel()    # healthy bearing
signal_fault   = data['X130_DE_time'].ravel()    # outer-race fault

print(f"Normal: {signal_normal.shape}  Fault: {signal_fault.shape}")
print(f"Sampling rate: 48 kHz | Duration: {len(signal_normal)/48000:.2f}s")

# — Cell 3: Visual exploration —————
fig, axes = plt.subplots(2, 1, figsize=(14, 6), sharex=True)

💡 Instructor cue: 'Look at the fault signal — the impulsive spikes every ~6.7 ms correspond to the outer-race ball-pass frequency. This is what RMS
alone misses but kurtosis catches.'
```

Step 2: Feature Engineering + Isolation Forest

STEP 2

```
# — Cell 4: Window-based statistical features —————
def extract_features(window):
    return pd.Series({
        'rms' : np.sqrt(np.mean(window**2)),
        'kurt' : kurtosis(window),
        'crest' : np.max(np.abs(window)) / np.sqrt(np.mean(window**2)),
        'skew' : skew(window),
        'std' : np.std(window),
        'peak2peak': window.max() - window.min(),
    })

W = 2048 # ~43 ms at 48 kHz
feats_norm = pd.DataFrame([extract_features(signal_normal[i:i+W])
                           for i in range(0, len(signal_normal)-W, W)])
feats_fault = pd.DataFrame([extract_features(signal_fault[i:i+W])
                            for i in range(0, len(signal_fault)-W, W)])
```



Discussion: Why train ONLY on 'normal'? What happens if we change contamination from 0.05 to 0.15? Trade-off between false alarms and missed faults.

```
iso = IsolationForest(contamination=0.05, n_estimators=200, random_state=42)
iso.fit(X_train_transform(feats_norm))
```

[1] Liu, Ting, Zhou (2008) IEEE ICDM — Isolation Forest [2] scikit-learn.ensemble.IsolationForest documentation

Step 3: Visualise & Interpret — From Score to Decision

STEP 3

```
# — Cell 6: Score distribution —————
plt.figure(figsize=(10, 4))
plt.hist(score_norm, bins=40, alpha=0.6, label='Healthy',
color='steelblue')
plt.hist(score_fault, bins=40, alpha=0.6, label='Faulty',
color='firebrick')
plt.axvline(iso.offset_, color='black', ls='--',
            label=f'Decision threshold ({iso.offset_:.3f})')
plt.xlabel('Anomaly score (lower = more anomalous)')
plt.ylabel('Window count'); plt.legend(); plt.tight_layout();
plt.show()

# — Cell 7: Confusion-style summary —————
from sklearn.metrics import classification_report
y_true = np.r_[np.zeros(len(feats_norm)), np.ones(len(feats_fault))]
y_pred = np.r_[(score_norm < iso.offset_).astype(int),
               (score_fault < iso.offset_) .astype(int)]
```

Group Discussion

Q
1

Which feature (RMS, kurtosis, crest) shows the biggest separation between healthy and faulty?

Q
2

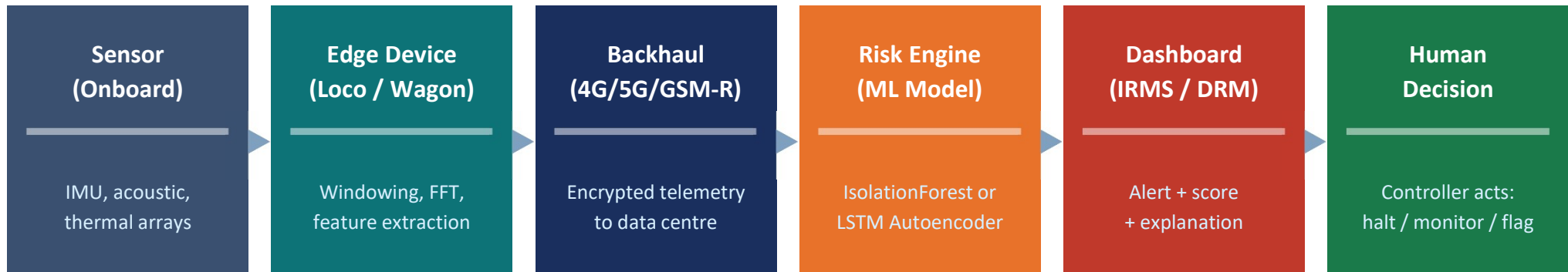
What is the cost asymmetry of a False Negative vs. a False Positive in a safety-critical system?

Q
3

At what threshold would you, as IRMS officer, escalate to a speed restriction or halt order?

💡 Stretch challenge: Replace IsolationForest with OneClassSVM ($\gamma='auto'$, $v=0.05$). Compare precision/recall. Which would you deploy on a freight rake? Defend your choice.

From Notebook to Field — A Production Pipeline



Indian Railways Deployment Anchors

CRIS e-Drishti / RailDrishti

Existing dashboard with asset-failure, bridge & track-renewal modules — natural surface for AI risk overlays¹

WILD + OMRS

24 WILD + 25 OMRS systems already deployed; 3 MVIS on NF Rly + DFCCIL — pilot foundation for fleet roll-out³

Kavach (TCAS) SIL-4

Commissioned on 1,452 RKM (Mar 2026); 4,154 locos fitted; ₹2,763.9 Cr cumulative spend²

Open data on data.gov.in

Period-wise consequential accident series available 2004-05 to 2023-24⁴

[1] CRIS e-Drishti project page [2] DD News, Mar 2026 (MoR Lok Sabha reply) [3] IBEF citing MoR (2024) [4] data.gov.in / Rajya Sabha Q.2195

SECTION 3

Open discussion: defining the operational framework for IRMS officers when AI flags a high-probability safety event

45 MINUTES

The 'Human-in-the-Loop' Requirement

THE FOUNDATIONAL INSIGHT

"Ironies of Automation"

Lisette Bainbridge, 1983¹

2,365 citations as of May 2026 — the paper's relevance is still rising. The two core ironies:

- (a) Automation removes the routine skills the operator needs for the rare intervention.
- (b) Operators must be MORE — not less — trained for abnormal conditions.

Implications for IRMS Safety AI

Alarm fatigue is real

ICU studies (Chambrin 1999): only 27% of monitoring alarms had clinical relevance². Surgical-monitor false-alarm rates: 85–99%.

Skills atrophy

Loco pilots who never manually intervene lose the heuristic ability to detect 'something doesn't feel right' — exactly the skill needed when AI fails.

Accountability gap

If AI halts a Rajdhani and causes commercial loss — who is liable? IR rules require human accountability for SIL-4 systems³.

[1] Bainbridge, L. (1983). Ironies of automation. *Automatica* 19(6), 775-779 [2] Chambrin et al. (1999) *Intensive Care Med.* PMID 10660842 [3] EN 50129; Kavach SIL-4 certification (RDSO)

Operational Framework — When AI Flags a High-Probability Event

1. ALERT

AI flags: anomaly score crosses threshold on Wagon W-7, Axle B of Train 12952. Risk = 87%.

2. VERIFY

Controller checks: Is the sensor healthy? Are corroborating channels (temp, GPS, time-since-last-maintenance) consistent?

3. CONTEXT

Add field-level info: Is this a known rough section? Is there an active monsoon advisory? Has this axle been flagged before?

4. DECIDE

Choose: HALT • Speed restriction to 30 kmph • Send guard for inspection • Dismiss with logged justification.

5. FEEDBACK

Log decision + actual outcome → feeds back into model retraining + governance audit trail (required by EN 50128).

Multi-Layered Governance Framework

Layer	AI Responsibility	Human Responsibility	Standard / Reference
Sensor / Edge	Real-time anomaly flagging with score	Sensor calibration, fault tolerance	RDSO HAHW spec, ISO 10816 ¹
Divisional Control	Rank risk, suggest action	Halt / speed restriction decision; incident log	IRMS rule book, EN 50129 ²
Zonal HQ	Model performance dashboards, drift detection	Policy decisions; threshold approval	EN 50128 verification, Model Cards ³
Railway Board / MoR	Audit trail, explainability reports	Regulatory compliance, AI procurement policy	EU AI Act Annex III analogue ⁴ ; DPDPA 2023
RDSO / Tech Wing	Model versioning, re-training protocols	Sensor specs, standards	EN 50126 RAMS lifecycle ²
External / Citizens	Transparent disclosure (Model Cards)	Right to grievance via RailMadad	NITI Aayog Responsible AI ⁵

[1] ISO 10816/20816; RDSO HAHW [2] EN 50126/50128/50129 functional safety stack [3] Mitchell et al. (2019) Model Cards, FAT* [4] EU AI Act Annex III Part 2 (high-risk: rail safety components) [5] NITI Aayog (2021) Responsible AI

OPEN DISCUSSION

THE SCENARIO

03:17 hrs. Train 12952 Rajdhani Express is 180 km into its journey. Onboard WADS generates an alert: Wagon W-7 Axle B. Kurtosis = 14.7 (baseline 3.2). Anomaly score: 0.91 (threshold 0.75). Axle-box temperature: 68°C (baseline 42°C). No other axles anomalous. Train on time. Next halt: 65 km. Single-line section. Dense traffic block.

YOUR TASKS (15 minutes, groups of 4)

1. Interpret: What do the kurtosis value and temperature rise tell you? Confirmed fault or risk signal only?
2. Decide: Halt immediately? Restrict to 30 kmph? Continue to next station? Defend your call.
3. Govern: Whom do you notify? What do you log? What is your legal exposure under EN 50129 / IR safety rules?

This is exactly the kind of decision the 'Ironies of Automation' anticipated. Defend your reasoning, not just your conclusion.

KEY TAKEAWAYS

1

Paradigm Shift:

Move analytics from post-incident forensics to continuous, sensor-driven risk scoring. CBM is the foundation — and the data already exists on IR.

2

Sensors & Features:

Vibration (kurtosis), temperature (Δ from baseline), and acoustic signals are your primary data streams. Features matter more than model choice.

3

Models Are Tools, Not Oracles:

Isolation Forest, OC-SVM, LSTM-AE — all have FP/FN rates. Calibrate thresholds to your operational context, not vendor brochures.

4

Human Judgement is the Final Check:

Bainbridge 1983 still holds. Automated alerts augment, not replace, the IRMS officer's accountability and field context.

5

Governance from Day One:

Define who owns the model, who reviews alerts, who logs decisions, how retraining is triggered. Build BEFORE deployment, not after.

References (1/2) — Sources Cited in This Session

REFERENCES

1. CAG (2022). Performance Audit on Derailment in Indian Railways. Report No. 22 of 2022, Union Government (Railways). cag.gov.in
2. Ministry of Railways via DD News (Mar 2026). 'Kavach 4.0 commissioned on 1,452 km...' Lok Sabha reply by Minister Ashwini Vaishnaw.
3. PIB Ministry of Railways (Jul 2023). 'Steep Decline in Consequential Train Accidents from 473 in 2000-01 to 48 in 2022-23.' Release ID 1941440.
4. data.gov.in (2024). Period-wise Consequential Train Accidents 2004-05 to 2023-24. Rajya Sabha Unstarred Q. 2195.
5. Commissioner of Railway Safety findings on Balasore (2 Jun 2023), reported in The Tribune (Jul 2023) and PMC12039713.
6. PRS India (2025). Demand for Grants 2025-26 Analysis: Railways. prsindia.org
7. ERA (2025). Safety Overview 2025. European Union Agency for Railways. era.europa.eu
8. UK ORR (2025). Rail Safety 2024-25. dataportal.orr.gov.uk
9. Association of American Railroads (2025). FRA Data Confirms Rail Safety Progress. aar.org
10. McKinsey (2017). The Rail Sector's Changing Maintenance Game. mckinsey.com
11. Deloitte Insights (2017). Industry 4.0 and Predictive Technologies for Asset Maintenance. deloitte.com
12. UIC (Feb 2024). The Journey Toward AI-Enabled Railway Companies. uic.org
13. OpenText (2017). Knorr-Bremse Group case study (D. Seckler quote on 20% cost reduction, 2-4 yr ROI). opentext.com
14. Knorr-Bremse Newsroom (Dec 2023). Digital trains: Knorr-Bremse and Nexxiot present connectivity solutions.
15. Alstom (2023). HealthHub & FlexCare product pages. alstom.com
16. Siemens Mobility. Railigent X & Velaro brochures. mobility.siemens.com

References (2/2) — Methods, Standards, Governance

REFERENCES

17. Liu, F.T., Ting, K.M., Zhou, Z.-H. (2008). Isolation Forest. Proceedings of the 8th IEEE ICDM, pp. 413-422. DOI: 10.1109/ICDM.2008.17
18. Schölkopf, B., Platt, J.C., Shawe-Taylor, J., Smola, A.J., Williamson, R.C. (2001). Estimating the Support of a High-Dimensional Distribution. Neural Computation 13(7), 1443-1471.
19. Malhotra, P., Ramakrishnan, A., Anand, G., Vig, L., Agarwal, P., Shroff, G. (2016). LSTM-based Encoder-Decoder for Multi-Sensor Anomaly Detection. ICML 2016 Workshop. arXiv:1607.00148. (TCS Research India)
20. Christ, M., Braun, N., Neuffer, J., Kempa-Liehr, A.W. (2018). tsfresh — Time Series Feature Extraction. Neurocomputing 307, 72-77.
21. Mitchell, M. et al. (2019). Model Cards for Model Reporting. ACM FAT* 2019, pp. 220-229.
22. CWRU Bearing Data Center, Case School of Engineering. engineering.case.edu/bearingdatacenter
23. Bainbridge, L. (1983). Ironies of automation. Automatica 19(6), 775-779. DOI: 10.1016/0005-1098(83)90046-8
24. Chambrin et al. (1999). Multicentric study of monitoring alarms in the adult ICU. Intensive Care Med. 25(12), 1360-1366. PMID 10660842.
25. EU Artificial Intelligence Act, Annex III Part 2 — high-risk AI systems in critical infrastructure. artificialintelligenceact.eu
26. CENELEC EN 50126 (RAMS lifecycle), EN 50128 / EN 50716 (software), EN 50129 (safety case for signalling).
27. NITI Aayog (2021). Responsible AI #AIForAll — Parts 1 (Principles) & 2 (Operationalising). niti.gov.in
28. RDSO. Specification for Hot Axlebox / Hot-Cold Wheel Detector System. RDSO-SPN-RE-HAHW-2012, rev. 2018.
29. ISO 10816 / ISO 20816 — Mechanical vibration evaluation of machine vibration.
30. TSB Canada (2014). Railway Investigation Report R13D0054 — Lac-Mégantic.

Questions & Discussion

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IIM Kozhikode MDP • May 2026

